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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S4 (S, FE) / S2 (PT) (S, FE) / S4 (WP) (S) Examination December 2024 (2019 Scheme)

Course Code: EET 206**Course Name: DIGITAL ELECTRONICS**

Max. Marks: 100

Duration: 3 Hours

PART A*(Answer all questions; each question carries 3 marks)*

Marks

- 1 Subtract 32 from 21 using 8-bit 2's complement arithmetic. Write the necessary steps. (3)
- 2 Compare the performance of TTL and CMOS logic families. (3)
- 3 Express the Boolean function, $f(X, Y, Z) = XY + \bar{X}Z$ as product of maxterms. (3)
- 4 Derive an expression for the sum of a full adder. (3)
- 5 Implement the sum of minterms expression, $f(A, B, C) = \sum m(0, 1, 5, 6)$ using a 4:1 multiplexer. (3)
- 6 List three functions of Arithmetic and Logic Unit (ALU). (3)
- 7 Derive the characteristic equation of a J-K flip-flop. (3)
- 8 What is meant by 'race around' in flip-flops? How it can be eliminated? (3)
- 9 Draw the circuit diagram of a 4-bit weighted resistor type digital to analog converter (DAC). Write an expression for its output. (3)
- 10 Define any three performance parameters of an analog to digital converter (ADC). (3)

PART B*(Answer one full question from each module, each question carries 14 marks)***Module -1**

- 11 a) Perform the following number conversions: (6)
 - (i) $(476.37)_8$ to binary.
 - (ii) $(A9C.FB)_{16}$ to octal.
 - (iii) $(101110.101)_2$ to decimal.
- b) Explain CMOS NOR gate with the help of internal diagram. (8)
- 12 a) Find out equivalent decimal number if 1101101 is expressed in (i) sign-magnitude form (ii) 1's complement form (iii) 2's complement form. (6)
- b) With the help of a neat internal diagram, explain the working of TTL NAND gate. (8)

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Module -2

- 13 a) With the help of a truth table, derive an expression for the difference and borrow bits of a full subtractor. Draw the logic diagram. (8)
- b) Simplify the Boolean function $f(A, B, C, D) = \sum m(1, 4, 5, 6, 7, 10, 12) + d(8, 11, 13)$ (6)
- 14 a) Simplify the Boolean functions to minimum number of literals: (6)
- (i) $f(P, Q, R, S) = P + Q[PR + (Q + \bar{R})S]$
- (ii) $f(X, Y, Z) = \overline{(X + \bar{Y}Z)} (X\bar{Y} + XYZ)$
- b) With a neat circuit diagram explain the working of a 4-bit parallel adder. What is its drawback when compared to a carry look-ahead adder? (8)

Module -3

- 15 a) Write down the truth table of a 2 to 4 decoder with active-low enable input and active-high outputs. Implement this decoder using AND and NOT gates. (5)
- b) Design and implement a 4-bit binary to Gray code converter. (9)
- 16 Write down the truth table for a BCD-to-seven segment decoder. Deduce the Boolean expression for each output. (14)

Module -4

- 17 a) Convert an S-R flip-flop to a J-K flip flop. Write the necessary truth table. (6)
- b) What is a shift register? Draw the logic diagram of a 4-bit serial-in, serial-out (SISO) shift register using D flip-flops. Explain its operation with the help of an example. (8)
- 18 a) Draw the logic circuit diagram of a 4-bit ring counter which employs D flip-flop. Write down its count sequence table. (5)
- b) Design a mod-10 asynchronous up-counter using J-K flip-flops. Draw the timing diagram and relevant excitation table. (9)

Module -5

- 19 a) What is PAL? List three advantages of PAL over PLA. (5)
- b) Draw the block diagram of a successive approximation type ADC. Explain how it converts an analog signal to a digital signal. (9)
- 20 a) List any four applications of FPGA. (4)
- b) Implement a full adder in Verilog. Draw the necessary logic circuit and truth table. (10)
